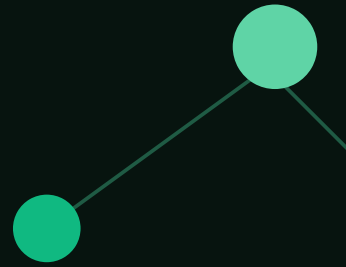


SymbioLink AI

A digital industrial-symbiosis platform that turns one factory's waste into another factory's cheap raw material.



TEAM

Team CircuLink

R.M.D Engineering College

1. Vishwa
2. _____
3. _____
4. _____

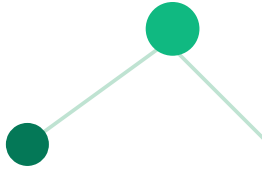
FACULTY MENTOR

PROJECT AT A GLANCE

Working prototype • Flask + MySQL • 3 trained ML models
Multi-hop waste-to-resource matching across an industrial cluster

THE PROBLEM

Usable industrial waste is going to landfill while MSMEs pay full price for raw material



Small and medium manufacturing units (MSMEs) inside a shared industrial cluster routinely generate waste that could serve as a nearby unit's raw material — but no digital system exists to discover, verify, and act on that match today.

EXISTING CHALLENGES

1 Manual discovery only

Matches happen through personal networks and word of mouth, so most usable waste is simply missed.

2 Multi-step exchanges are invisible

One unit's waste feeding another's byproduct feeding a third can't be seen by a simple listings board.

3 Literacy & app-adoption barriers

Form-heavy, app-only platforms shut out many MSME owners who would otherwise take part.

4 No trust signal

There is no way to check a counterpart's reliability before committing to a physical exchange.

OUR SOLUTION

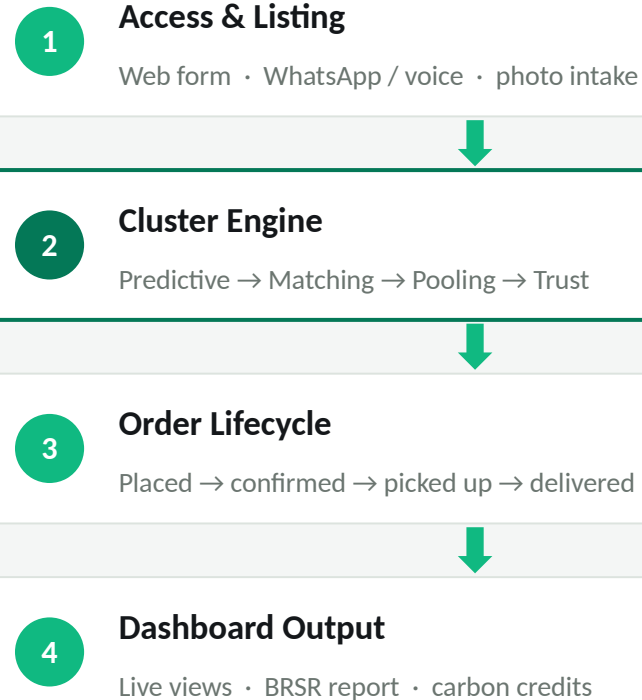
SymbioLink AI: a graph of the whole cluster, not a one-to-one classifieds board

The cluster is modeled as a directed graph. The engine searches for direct matches AND 2-3 hop exchange chains (Unit A → B → C), each ranked by estimated cost savings, transport distance, and a trust score built from past completed exchanges.

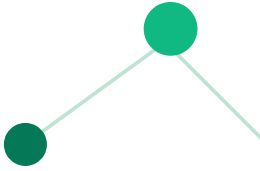
WORKING METHODOLOGY

- 1 Listings via web form, or a WhatsApp / voice-note message parsed into structured data
- 2 Predictive layer forecasts a unit's next waste listing before it's even generated
- 3 Logistics pooling groups small loads into shared transport runs
- 4 Order lifecycle feeds a two-way trust rating back into future match rankings

SYSTEM WORKFLOW



Four trained models replace what used to be hand-coded heuristics — each still explainable



1

Intake Classifier

TF-IDF + Logistic Regression

Parses WhatsApp / voice-note text (incl. Hinglish) into a structured listing

2

Forecast Regressor

Ridge Regression

Predicts a unit's next likely waste listing from its posting history

3

Match Ranker

Logistic Regression

Scores each chain on saving, distance, trust, similarity and hop count

4

Photo Classifier

Random Forest

Guesses material family and quantity from an uploaded waste photo

DATASETS

Synthetic dataset modeled on documented Indian cluster patterns (Muzaffarnagar paper/sugar-mill belt, Mysore mixed-manufacturing) — 12 MSME units, ~20 listings, reference price table, Hinglish phrasing examples, and procedurally generated material-family images.

TECHNOLOGY STACK

Python 3

Flask

MySQL (PyMySQL)

scikit-learn

Google OR-Tools

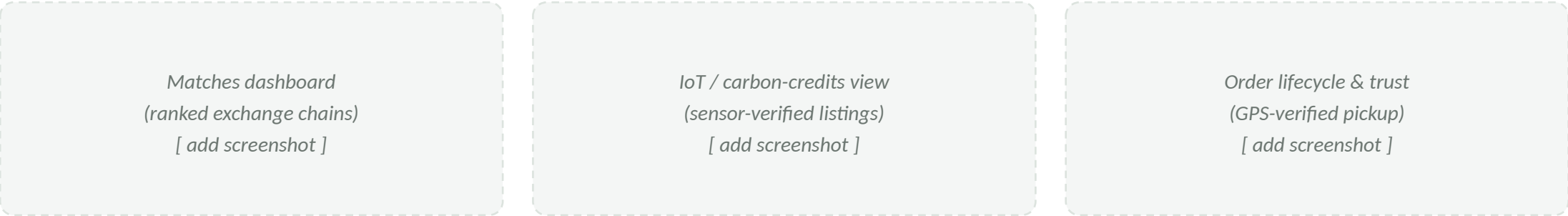
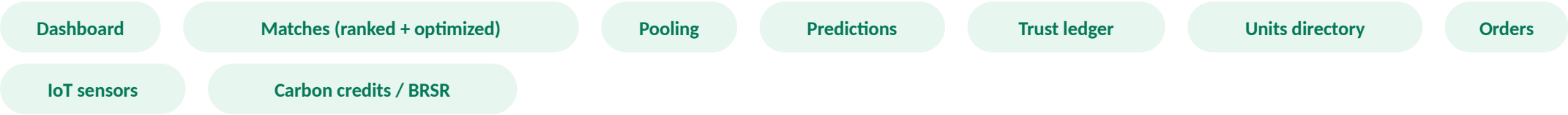
Pillow / NumPy

Twilio WhatsApp

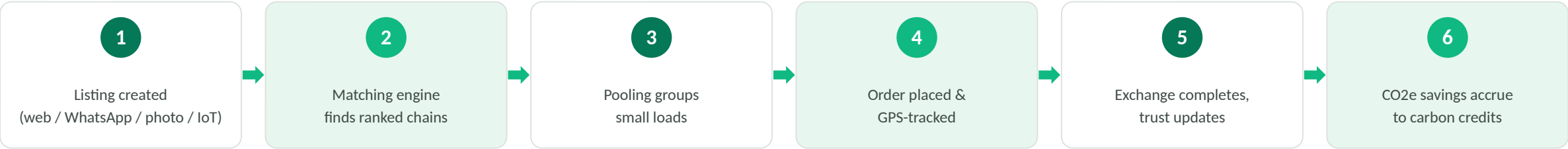
Google Maps API

PROTOTYPE

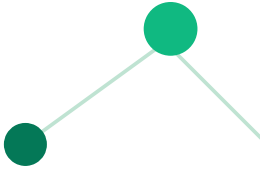
A working Flask dashboard — nine live pages tying every layer together



PROCESS FLOW



Multi-hop, predictive, and explainable — not a one-to-one classifieds board



	EXISTING SOLUTIONS	SYMBIOLINK AI
Matching depth	✗ One-to-one listings only	✓ Direct matches AND 2-3 hop exchange chains (A → B → C)
Timing	✗ Reactive — posted after waste exists	✓ Predictive layer forecasts listings before waste is generated
Intake	✗ Text-only, app-required forms	✓ WhatsApp / voice-note + photo-based intake, Hinglish-aware
Verification	✗ No counterpart verification	✓ Two-way trust ledger + GPS-verified pickup/delivery
ESG linkage	✗ No compliance or carbon tie-in	✓ Built-in BRSR compliance report + carbon-credit marketplace

Uniqueness: every AI-ranked score still returns a plain-language reason (explain_match()) naming the strongest factor behind it — transparent by design, not a black box.

EXPECTED IMPACT

Cheaper inputs for MSMEs, less to landfill, auditable environmental data for the cluster



Raw-material cost

Lower input cost via nearby waste-as-resource matches



Landfill diversion

Usable waste kept in productive use across the cluster

CO₂e

Verified savings

Logged per exchange, convertible into tradable credits

ESG

Audit-ready reporting

BRSR Principle 6 line items generated from real order history

TARGET USERS

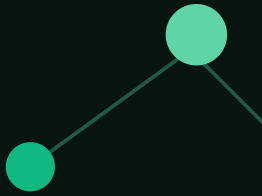
MSME owners inside manufacturing clusters · industrial cluster associations · sustainability / ESG officers coordinating compliance across member units

SCALABILITY

The rules-based graph engine plus trained ML layer generalizes to any documented industrial cluster — the prototype is built to onboard a real partner cluster association's data with no architecture changes, starting from Muzaffarnagar- and Mysore-style patterns.

ROADMAP

From working prototype to a real cluster pilot



CURRENT PROGRESS

Working Flask prototype · 6 core layers + 7 added modules (IoT, BRSR, photo classifier, OR-Tools optimization, Hinglish NLP, geofencing, carbon credits) · 3 trained ML models · MySQL persistence · automated test suite

FUTURE ROADMAP

1 Onboard one real partner cluster association's listings

2 Wire up the Twilio WhatsApp webhook for live intake

3 Move to managed/hosted MySQL for production

4 Tune price & transport-rate tables against real quotes

5 Flash real ESP32 IoT sensors; live vehicle GPS feed

6 Retrain the photo classifier on real waste photos

COMMERCIALISATION: pilot with a cluster association as anchor customer → register with a recognized voluntary carbon methodology/registry → SaaS / commission-per-exchange model